

Jingkun Qian

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EDUCATION

2024 – Present **BSc Artificial Intelligence and Computer Science**, University of Edinburgh (GPA: 85.8/100)

Selected Courses and Results:

- Introduction to Linear Algebra (MATH08057) — 92% (A1)
- Informatics 1: Object-Oriented Programming (INFR08029) — 93% (A1)
- Calculus and its Applications (MATH08058) — 87% (A2)

2021 – 2024 **GCE A-Level**, Beijing National Day School (GPA: 4.18/4.3)

PROJECTS

LLM-Integrated Neuro-Symbolic Reasoning System

Python, Large Language Models, Neuro-Symbolic AI, Function Calling, Agentic Systems

[GitHub Link](#)

- Designed and implemented a neuro-symbolic reasoning system integrating Google Gemini with the Stockfish engine, combining symbolic search with natural-language strategic analysis.
- Developed structured function-calling workflows enabling reliable interaction between large language models and external reasoning tools for position evaluation, conversational analysis, and decision support.

Automated Symbolic Planning System for Logistics Optimization

PDDL, Metric-FF, Automated Planning, Symbolic AI, Algorithm Benchmarking

Jan 2026 – Mar 2026

- Engineered a multi-agent domain and problem framework using PDDL incorporating numeric fluents for resource capacity limits, stamina depletion, and path distance costs.
- Benchmarked Metric-FF planner performance by evaluating search space expansions under variable heuristic weight adjustments (w_g and w_h).

Longitudinal Migration Analysis of Notable Individuals

Python, Data Science, Statistical Modelling, Logistic Regression, Geospatial Analysis

[GitHub Link](#)

- Conducted a large-scale longitudinal migration study on a Wikipedia–Wikidata dataset containing 165,000+ notable individuals, performing data cleaning, feature engineering, and geospatial mobility analysis across historical periods.
- Developed logistic regression models with robustness checks to investigate the effects of domain, birth cohort, gender, and notability on migration behaviour, reporting odds ratios, confidence intervals, and temporal mobility trends.

EXPERIENCE

Undergraduate Research Intern — Aerial Manipulation & Robotic Manipulation

Westlake University, School of Engineering | Isaac Sim, ROS 2, Robotic Manipulation, TAMP

Jun 2026 – Sep 2026

- Implemented an RViz InteractiveMarker tool for dragging the robot's calibration point, converting target poses to joint commands via inverse kinematics, and triggering replanning through ROS 2 topics
- Developed generalized robotic manipulation environments in NVIDIA Isaac Sim, including deformable-object simulation for flexible plastic bags and configurable scene randomization for diverse VLA manipulation data collection.
- Calibrated a quasi-static 6-DoF end-effector force estimation pipeline based on motor-current residuals, gravity compensation, and inverse kinematics, fitting torque and gravity scaling parameters against force-gauge measurements.

CERTIFICATES

- **CAIE Mathematics** — A*
- **CAIE Further Mathematics** — A*
- **CAIE Computer Science** — A*
- **USACO Bronze Division** — USA Computing Olympiad

SKILLS

Programming Languages: Java, C++, Python

Tools & Libraries: Git, NumPy, Pandas, MySQL, VS Code, IntelliJ IDEA, Anaconda